EXPERIMENT-24

24. Nmap Timing & Performance in Kali Linux

Aim

To execute Nmap timing and performance commands in Kali Linux using different timing templates and analyze the scanning speed and results of a target IP address.

Procedure

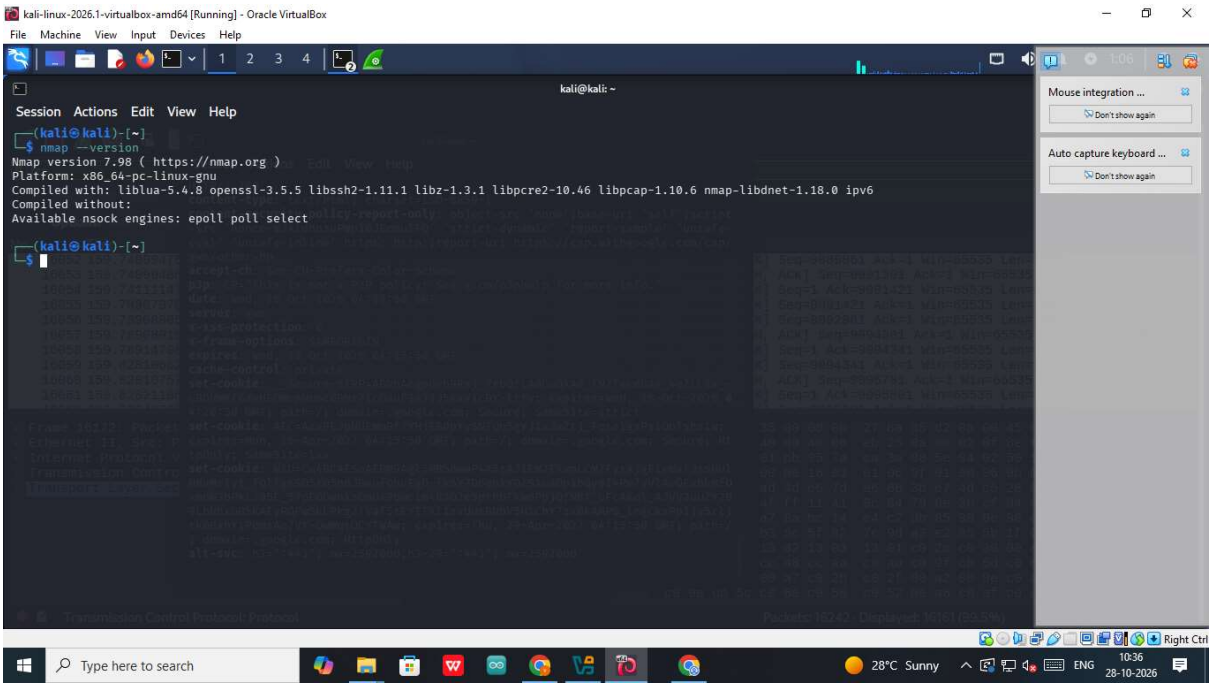
1. Start Kali Linux in Oracle VirtualBox.
2. Open the Terminal.
3. Check whether Nmap is installed.
4. Find the target machine's IP address.
5. Execute the Nmap command using the `-T3`, `-T4`, and `-T5` timing templates.
6. Observe the scan results and compare the speed and reliability of each scan.

Step 1: Check Nmap version

Input:

```
nmap --version
```

output:



Step 2: Find the IP address

To view your Kali Linux IP address, enter:

Input:

```
ip -br addr
```

output:



Step 3: Normal Timing Scan (-T3)

Input:

```
nmap -T3 192.168.1.10
```

Purpose: Performs a normal-speed scan using the T3 timing template.

output:

```
┌──(kali㉿kali)-[~]
└─$ nmap -T3 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-28 01:08 -0400
Nmap scan report for 192.168.1.10
Host is up (0.00050s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 8.16 seconds
```

Step 4: Aggressive Speed Scan (-T4)

Input:

```
nmap -T4 192.168.1.10
```

Purpose: Uses aggressive timing to complete the scan faster on a reliable network.

output:

```
┌──(kali㉿kali)-[~]
└─$ nmap -T4 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-28 01:10 -0400
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.50% done
Nmap scan report for 192.168.1.10
Host is up (0.00043s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.55 seconds
```

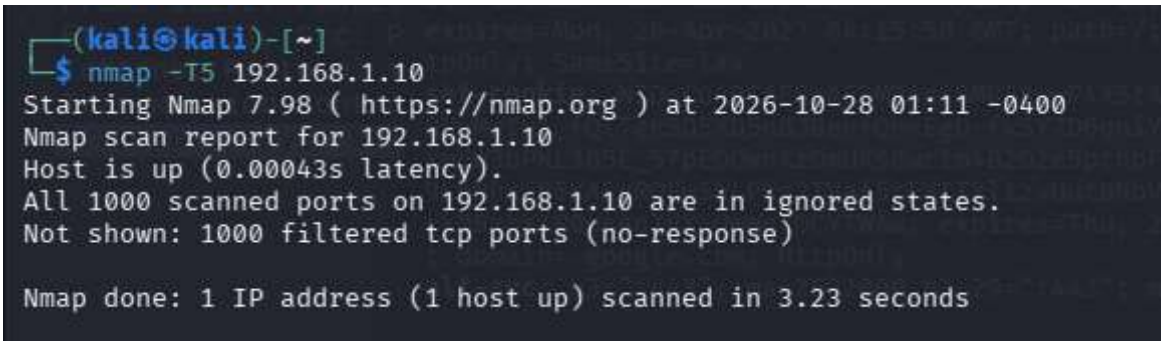
Step 5: Insane Speed Scan (-T5)

Input:

```
nmap -T5 192.168.1.10
```

Purpose: Uses the fastest and most aggressive Nmap timing template. It may miss results on slow or unreliable networks.

Sample output:



Result

Nmap timing and performance commands were executed to study normal, aggressive, and insane scan settings. The effects of different timing templates on scanning speed and reliability were compared.